

ZUERICH-AFFOLTERN, Switzerland

WMO: 066640

Lat: 47.4278N

Lon: 8.5178E

Elev: 1447

StdP: 13.94

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	20.6	11.2	10.3	19.2	15.5	12.7	22.9	24.6	46.9	20.9	47.0	3.6	50	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	87.9	68.0	84.2	66.8	80.9	65.8	69.7	83.1	68.3	80.5	66.9	77.9	4.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	99.6	73.7	64.2	94.9	72.3	63.0	90.8	71.0	34.5	83.2	33.3	80.5	32.2	78.4	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.3	14.2	11.6	DB	10.8	92.8	5.7	2.9	6.7	94.9	3.4	96.6	0.2	98.2	-3.9	100.3
			WB	10.0	73.1	6.0	1.6	5.7	74.3	2.2	75.2	-1.2	76.1	-5.6	77.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	34.1	35.1	42.4	50.0	57.1	64.9	67.4	66.4	58.9	51.2	41.5	35.2	(6)
	DBStd	13.74	7.41	7.63	7.13	6.48	6.45	6.44	5.65	5.67	5.83	6.69	6.40	6.78	(7)
	HDD50	2047	495	418	246	80	13	0	0	0	4	70	262	459	(8)
	HDD65	5626	959	837	700	451	255	81	40	51	196	429	704	923	(9)
	CDD50	2209	1	0	11	79	233	449	541	509	272	106	8	0	(10)
	CDD65	315	0	0	0	0	11	79	116	95	14	0	0	0	(11)
	CDH74	2960	0	0	0	15	156	778	1075	827	108	1	0	0	(12)
	CDH80	894	0	0	0	1	24	235	355	272	7	0	0	0	(13)
(14) Wind	WSAvg	3.7	4.5	4.5	4.6	4.0	3.8	3.4	3.2	2.9	2.8	2.8	3.7	4.4	(14)
(15) Precipitation	PrecAvg	47.5	2.7	2.8	3.0	3.5	5.2	5.4	5.5	5.2	3.8	3.5	3.5	3.5	(15)
	PrecMax	60.7	6.7	6.4	8.9	6.9	9.6	8.7	10.1	9.3	8.2	6.3	7.7	7.3	(16)
	PrecMin	37.4	0.4	0.8	0.9	0.5	1.2	2.0	2.1	0.7	1.8	0.3	0.0	0.0	(17)
	PrecStd	6.1	1.7	1.5	1.7	1.7	2.3	1.7	2.1	2.0	1.6	1.6	1.9	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	58.3	67.5	75.9	83.3	90.8	92.5	92.1	80.8	71.5	62.2	53.7	(19)
		MCWB	48.7	47.4	51.9	56.7	64.4	69.4	68.5	67.9	65.4	60.3	53.5	48.0	(20)
	2%	DB	51.4	53.3	63.2	71.6	78.4	86.2	87.9	87.0	77.6	67.3	57.2	50.5	(21)
		MCWB	46.0	45.1	49.8	55.0	62.2	67.7	67.7	68.3	64.3	58.6	51.6	46.5	(22)
	5%	DB	48.0	49.8	59.0	67.9	74.3	82.5	83.9	82.6	73.2	64.2	53.7	47.8	(23)
		MCWB	43.6	43.8	48.3	53.3	59.7	66.5	66.6	66.9	62.2	57.4	49.8	44.3	(24)
	10%	DB	44.7	46.0	55.0	63.7	70.3	78.4	80.2	78.3	69.6	60.9	50.8	45.2	(25)
		MCWB	41.4	41.1	46.8	51.8	58.1	64.8	65.5	65.8	61.0	55.8	47.4	42.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	48.5	53.5	58.5	65.9	72.0	71.6	71.2	67.5	62.0	54.5	49.4	(27)
		MCDB	53.8	55.1	64.5	72.5	79.5	86.1	86.5	83.9	77.5	68.4	60.1	52.2	(28)
	2%	WB	46.8	45.9	51.1	56.1	63.7	69.3	69.4	69.4	65.4	60.0	52.1	46.9	(29)
		MCDB	50.5	52.2	59.6	68.4	75.2	81.9	83.1	82.2	74.5	65.5	56.3	50.1	(30)
	5%	WB	43.8	43.7	49.3	54.4	61.4	67.5	67.9	67.9	63.5	58.2	50.1	44.8	(31)
		MCDB	47.6	49.4	57.4	65.4	70.9	79.1	80.1	79.7	71.2	63.0	53.6	47.6	(32)
	10%	WB	41.4	41.5	47.3	52.6	59.3	65.8	66.4	66.4	61.8	56.4	47.7	42.5	(33)
		MCDB	44.4	45.8	54.5	62.4	67.9	76.3	77.3	76.6	68.1	60.2	50.4	44.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.5	17.0	19.1	18.3	19.2	19.4	18.5	16.9	14.1	10.5	8.9	(35)
		MCDBR	14.1	20.4	25.9	28.3	26.7	26.4	27.0	26.3	22.8	19.3	15.7	12.0	(36)
	5% WB	MCWBR	9.6	12.9	14.3	13.7	12.1	10.3	10.0	10.0	10.9	11.2	10.4	8.7	(37)
		MCWBR	12.9	18.8	22.6	25.2	22.9	24.0	24.3	23.7	20.4	17.3	14.7	11.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.295	0.320	0.356	0.390	0.404	0.408	0.410	0.402	0.377	0.351	0.319	0.291	(40)
		taud	2.444	2.373	2.330	2.269	2.275	2.303	2.292	2.329	2.382	2.437	2.477	2.453	(41)
	Ebn at Noon		252	267	273	273	273	272	269	266	263	252	240	239	(42)
		Edh at Noon	21	28	34	40	41	40	40	37	32	26	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		371	638	1022	1375	1581	1815	1756	1533	1150	724	404	305	(44)
	RadStd		41	86	137	198	159	169	160	157	118	79	49	47	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)	N/A	N/A	N/A	+1.26	+0.76	+0.61	N/A	N/A	+101	+27

Nomenclature: See separate page